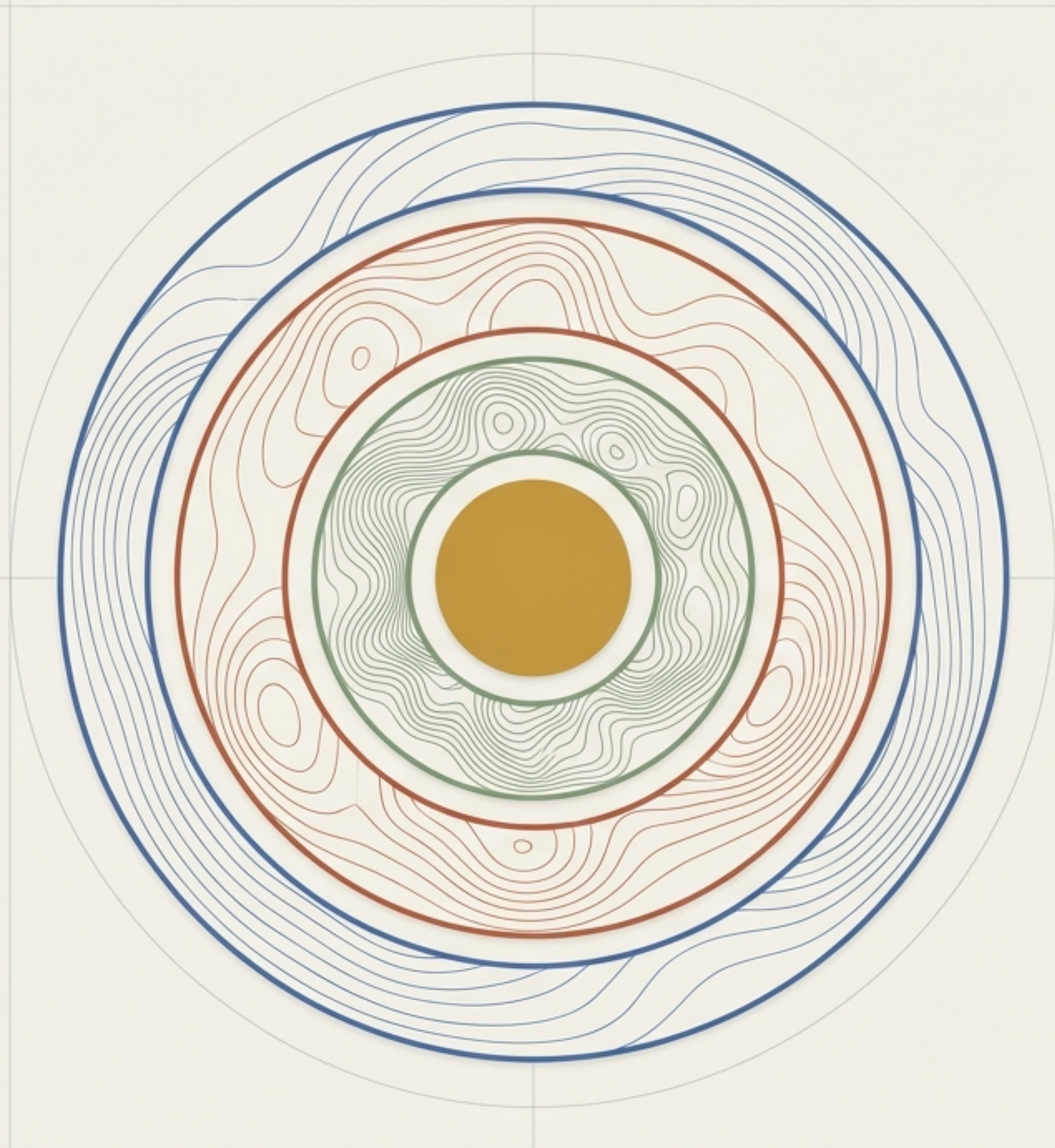


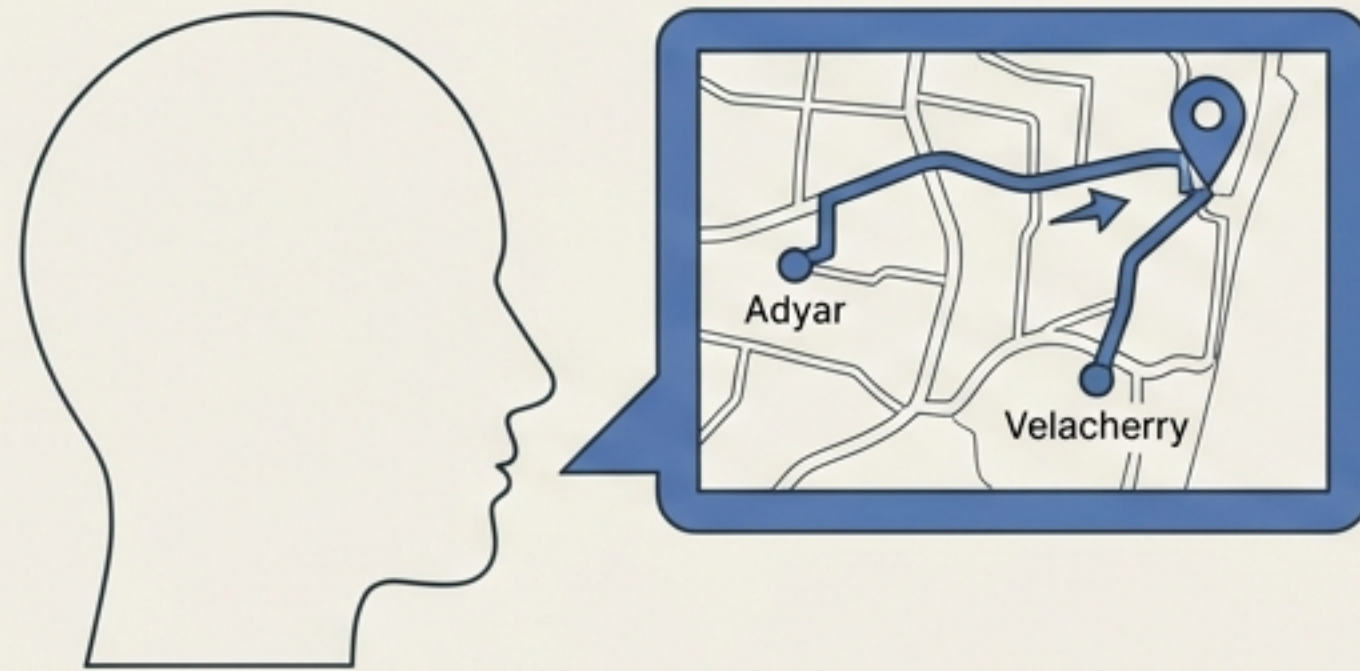
Decoding the Evolutionary Layers of Artificial Intelligence

Mapping the precise mechanics of how data transforms from structured localized predictions to multimodal generation across Machine Learning, Deep Learning, and GenAI.



AI mimics human behavior by marrying current scenarios with vast historical data to execute accurate predictions.

Human Context

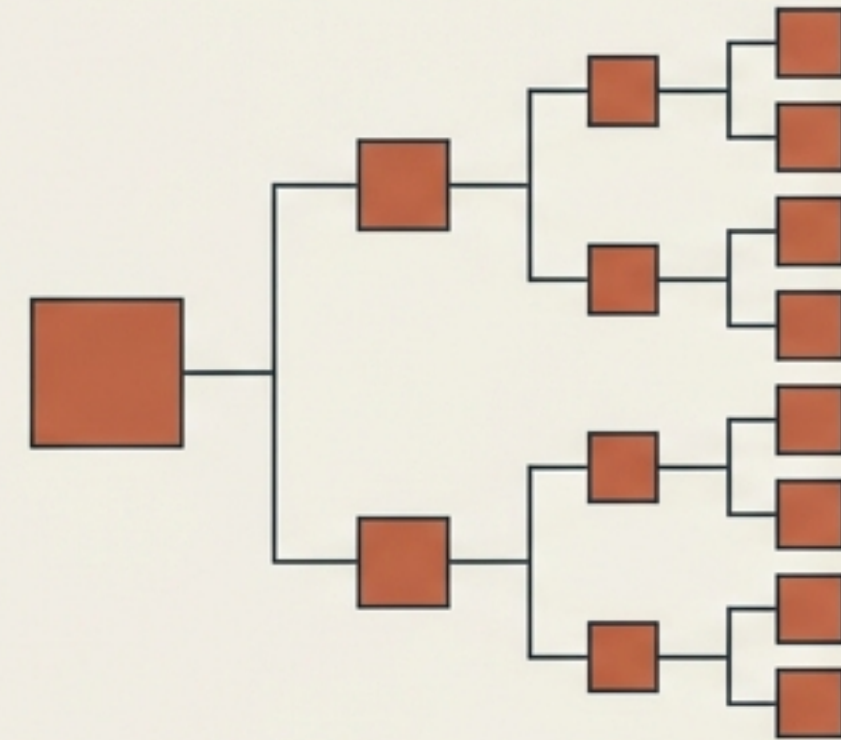


Use Case 1: Navigation

Challenge: Find the shortest route from Adyar to Velacherry at 9 AM in a car.

Mechanism: The AI processes millions of historical records mapping traffic density, distance, and transportation modes to predict precise travel time.

Algorithmic Context

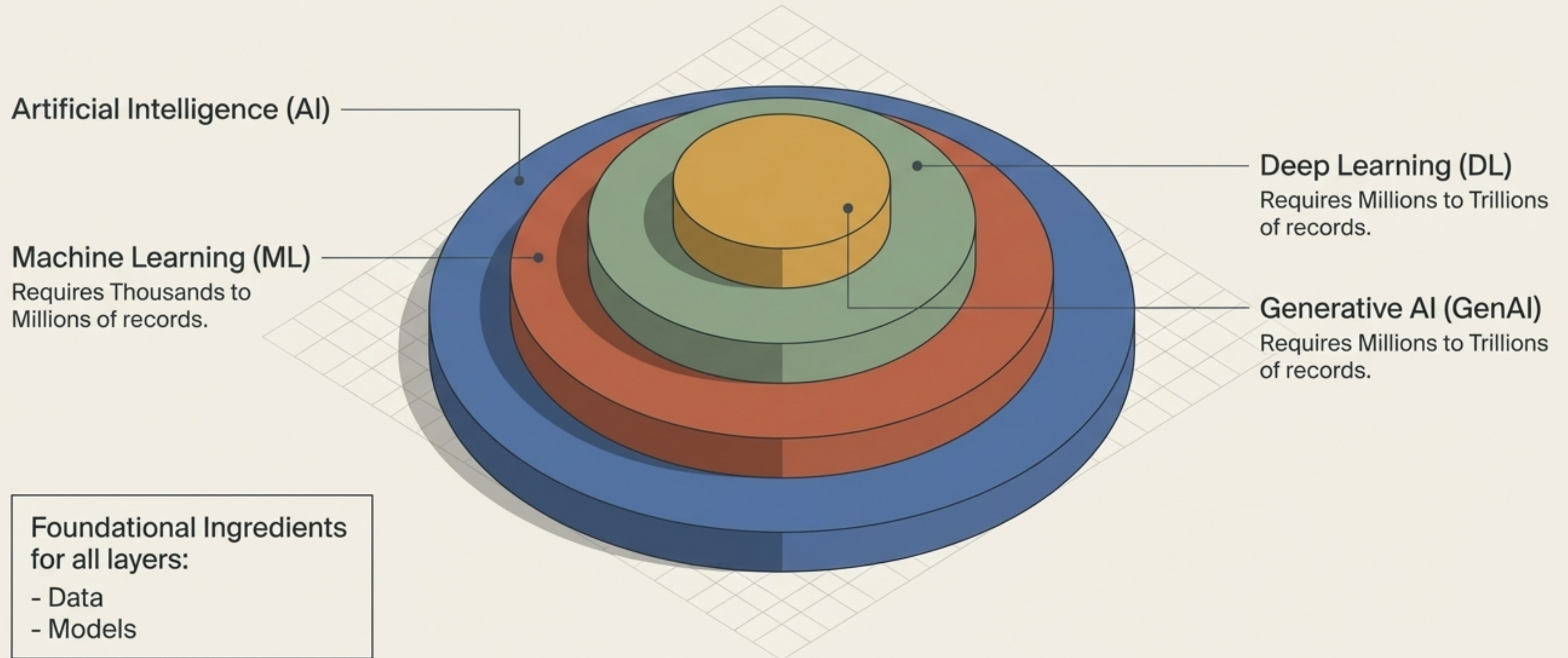


Use Case 2: Weather Forecasting

Challenge: Predict today's temperature to be 30 degrees.

Mechanism: The AI analyzes millions to billions of historical data points spanning the month of March across years like 1990, 2024, and 2025.

The AI ecosystem is a nested taxonomy defined by scaling data volumes and shifting operational mechanics.



Machine Learning extracts definitive predictive intelligence from localized datasets.

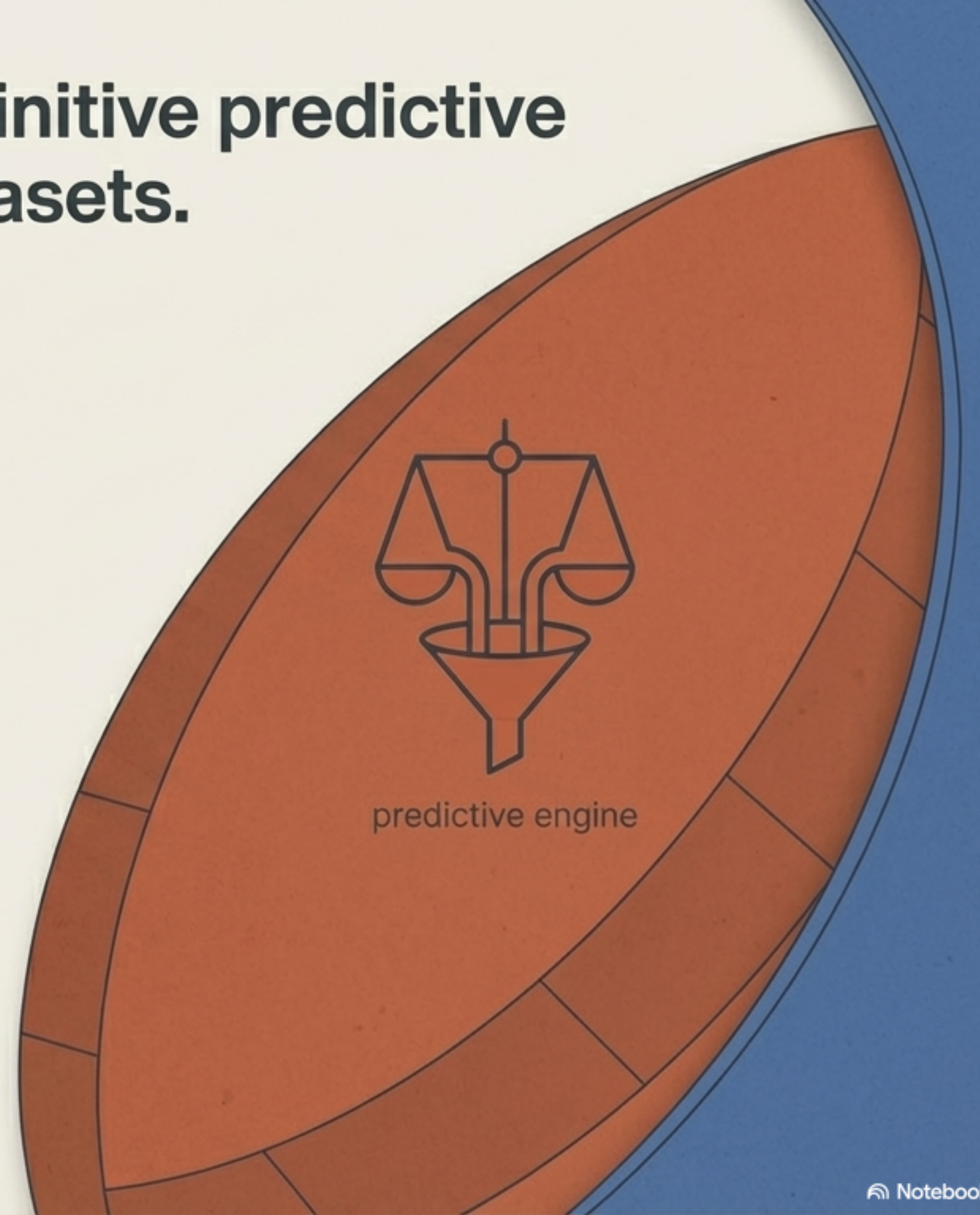
Definition:

The foundational subset of Artificial Intelligence.

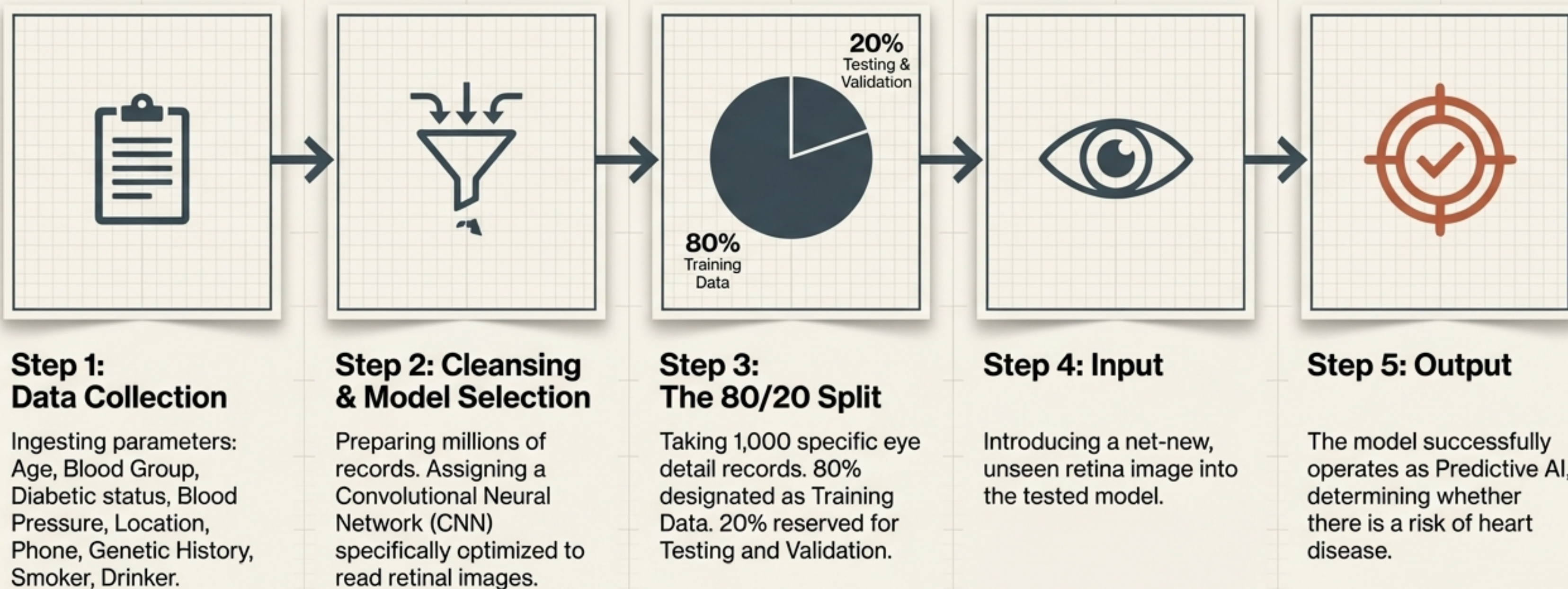
Data Threshold:

Operates on structured datasets ranging from 1,000 to millions of records.

“The definitive output of Machine Learning is entirely restricted to Prediction.”



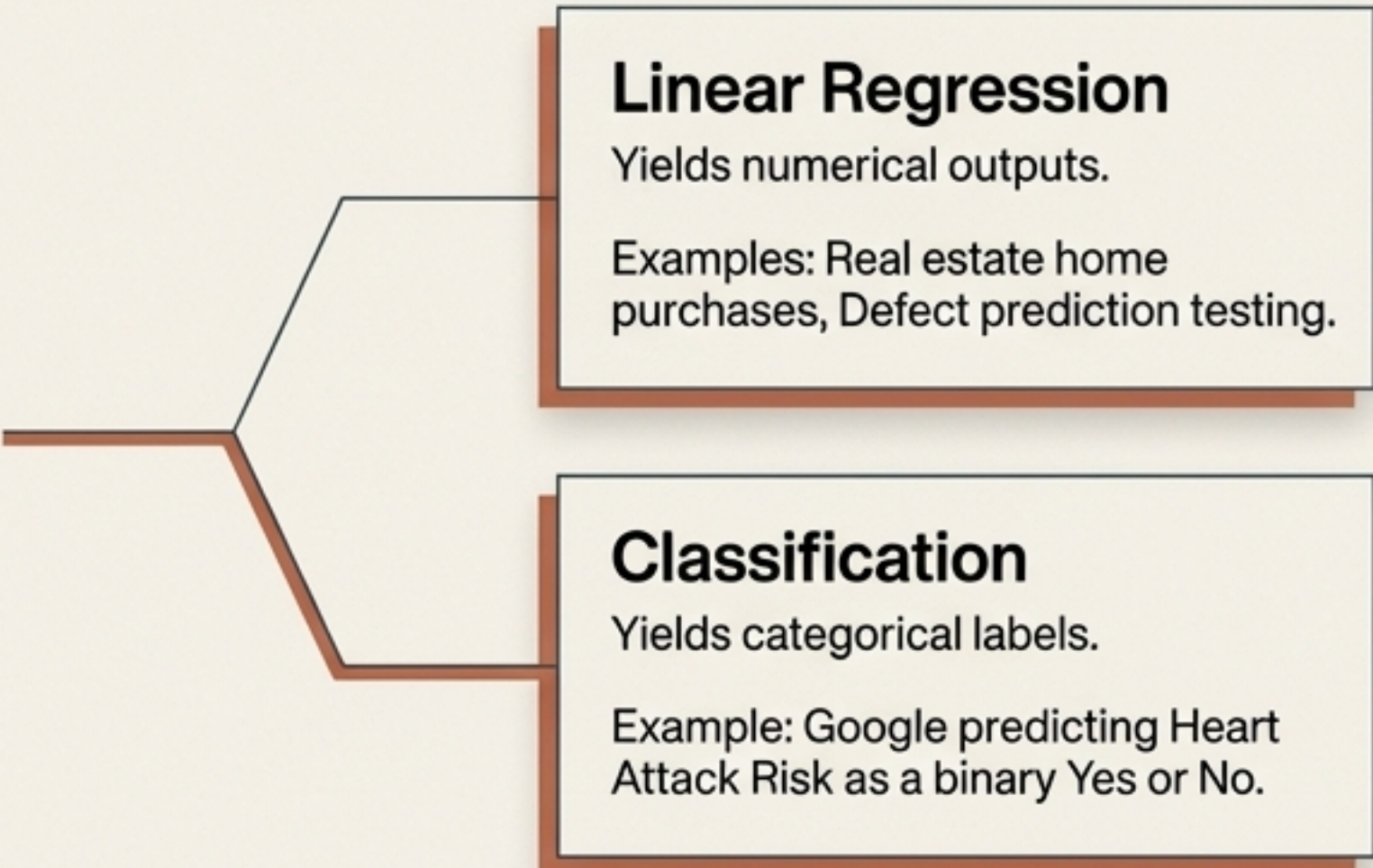
Training a predictive machine learning model requires structured data cleansing and rigorous validation splits.



Machine learning fractures into supervised data labeling and unsupervised behavioral pattern recognition.

Supervised Learning

Requires explicitly labelled data.



Linear Regression

Yields numerical outputs.

Examples: Real estate home purchases, Defect prediction testing.

Classification

Yields categorical labels.

Example: Google predicting Heart Attack Risk as a binary Yes or No.

Unsupervised Learning

Operates on unlabelled data based entirely on behaviors and genres.

Primary Methodology: Collaborative Filtering

Analyzes behavioral patterns to recommend content without relying on explicit user inputs like comments.

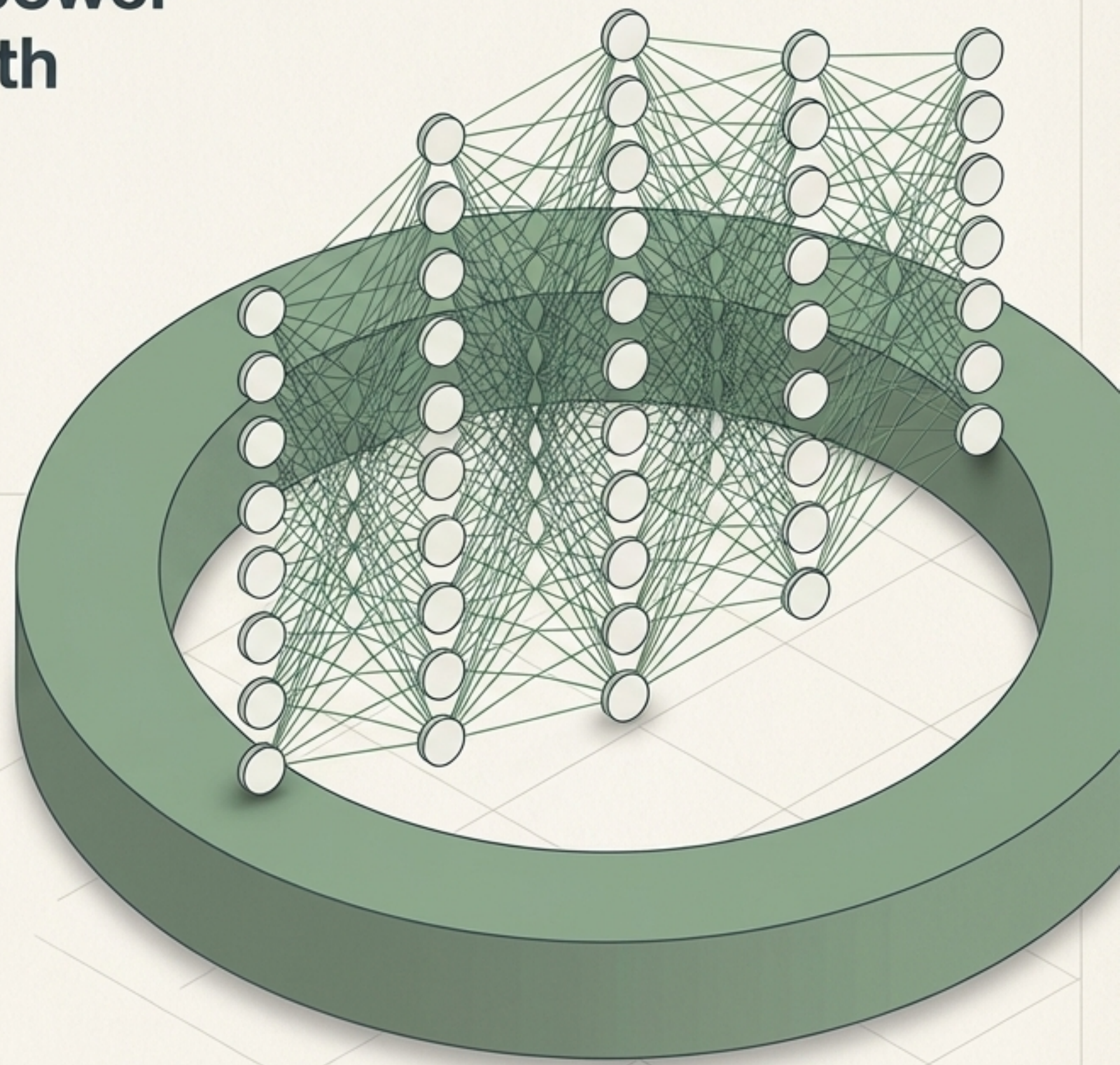
Warning

If a system relies on star ratings, likes, or dislikes, it reverts to Supervised Content-Based learning because it is using explicit labels.

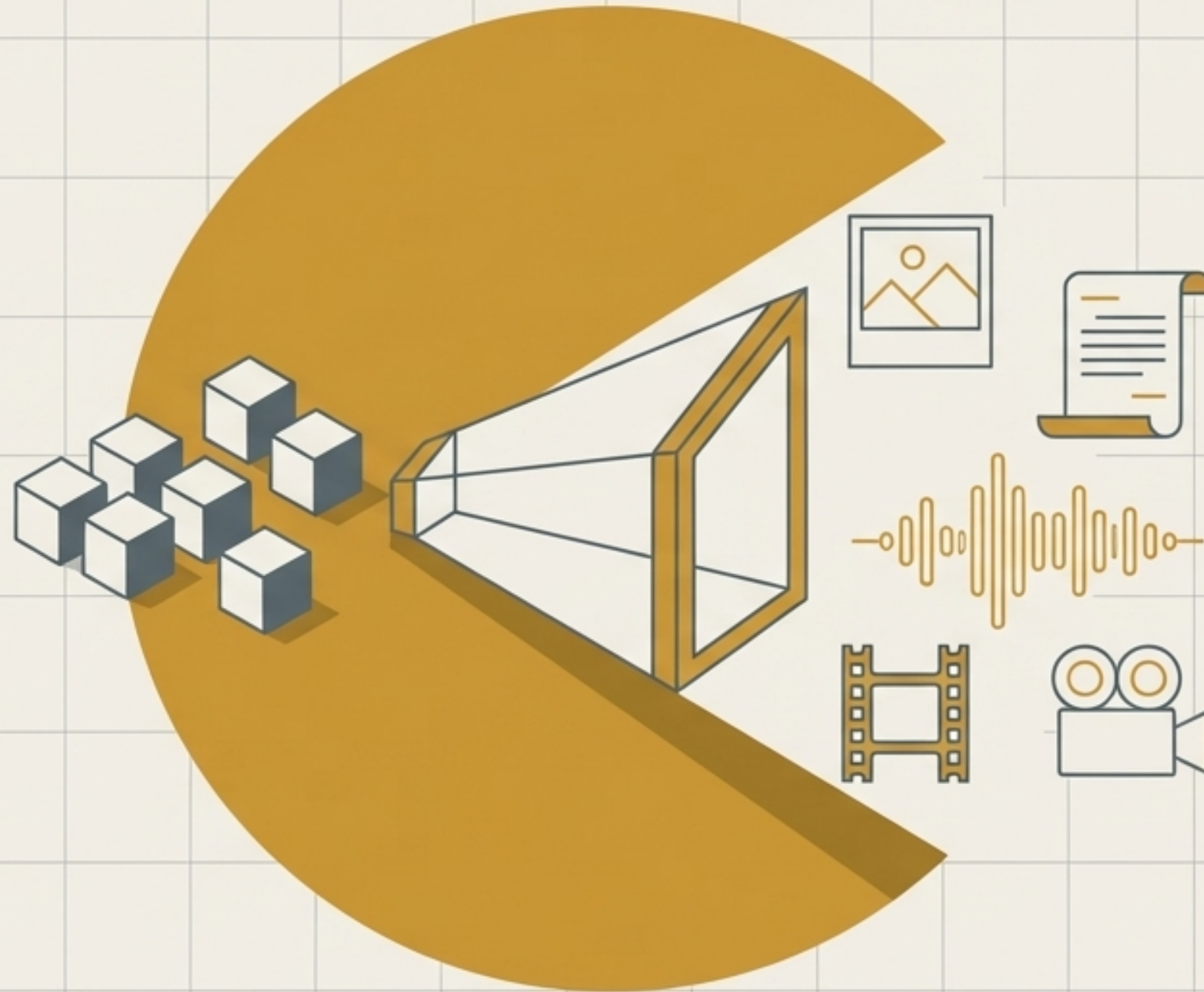
Deep Learning scales predictive power by marrying vast data volumes with complex neural networks.

Machine Learning + Vast Data
(Millions to Trillions of records)
+ Neural Networks
= Deep Learning

Narrative Anchor: Despite the immense jump in scale, density, and computational complexity, Deep Learning remains fundamentally a Predictive system.



Generative AI abandons the predictive model entirely to synthesize novel, multimodal content.



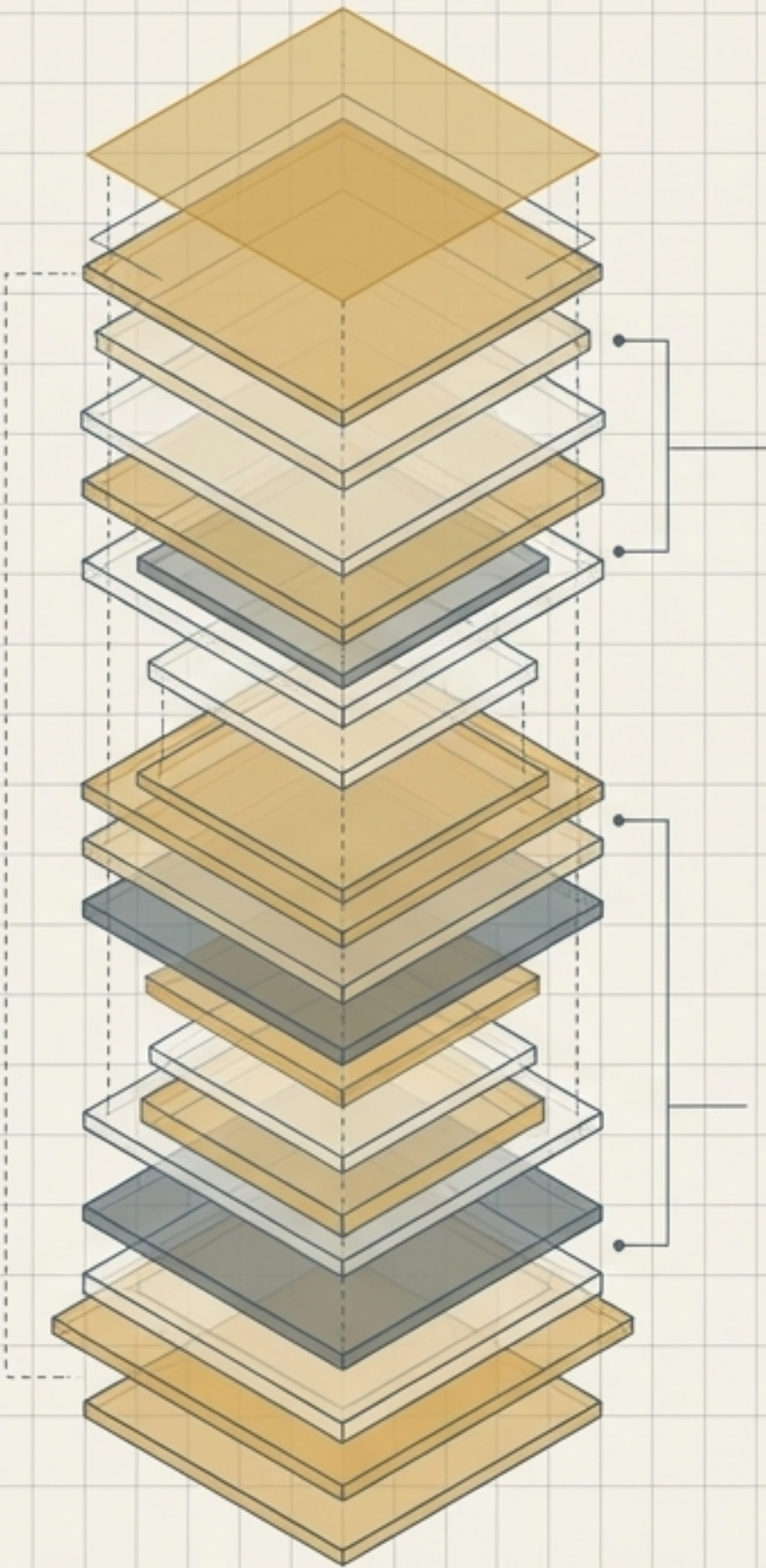
The Paradigm Pivot:

GenAI is a specialized subset of Deep Learning, but it breaks the predictive mold. It is NOT predictive.

Instead, it utilizes massive datasets spanning trillions of ubiquitous records to generate net-new, original content across all domains, including raw code.

MULTIMODAL

Generative Pre-trained Transformers (LLMs) function as the architectural engine beneath modern chat agents.



GENERAL SPECS	
Provider:	OpenAI
Primary Chat Agent:	ChatGPT
Core Architecture:	Large Language Model (LLM)

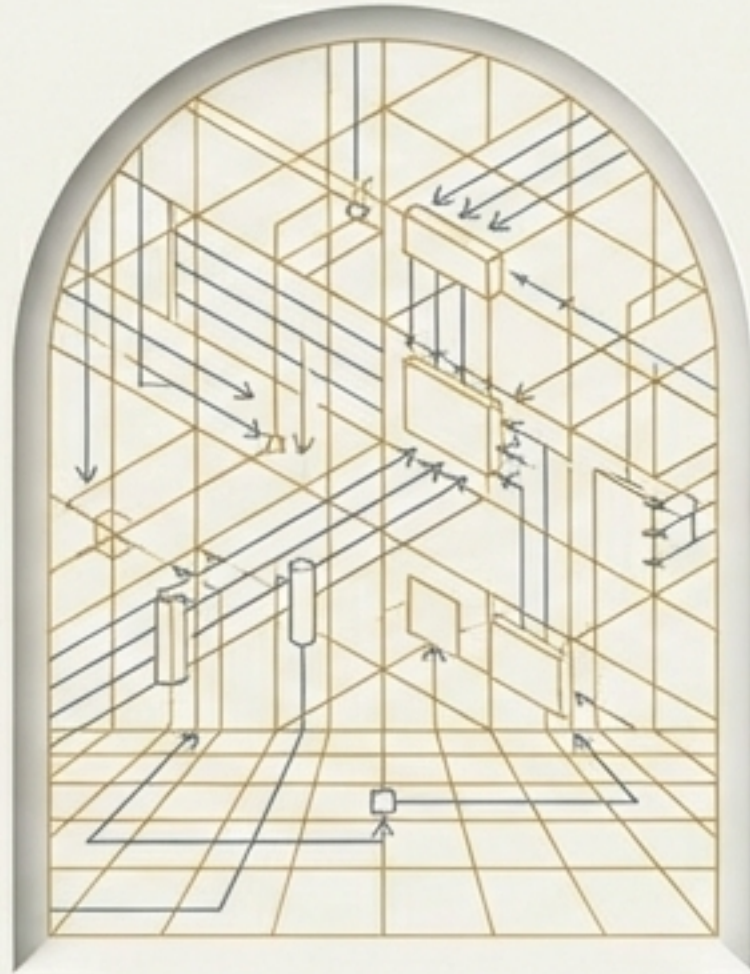
THE "P" IN GPT
Pre-Training Note: Pre-trained on huge, ubiquitous datasets spanning all fields and human knowledge, not limited exclusively to software engineering.

EVOLUTIONARY SPECIFICATIONS	
GPT-4	GPT-5
Architecture: 96 to 120 Processing Layers.	Architecture: Scales massively to 190 Layers. Capacity: 120 Billion Parameters. (Intermediate iterations include GPT 5.3 and GPT 5.4)

The Evolutionary AI Matrix: Tracing the shift from localized prediction to multimodal generation.

Layer	Core Function	Data Volume Threshold	Output Methodology	Defining Example
 Machine Learning (ML)	Predictive Function	1,000s to Millions of Records	Numerical or Categorical	Defect Prediction & Heart Attack Risk
 Deep Learning (DL)	Scaled Predictive Function	Millions to Trillions of Records	Neural Network Driven	Advanced Image Recognition (Retinal CNN)
 Generative AI (GenAI)	Generative Function	Trillions of Records (Ubiquitous)	Multimodal (Text/Image/Audio/Video)	Generating Novel Code via LLM

The ecosystem is mapped. Next, we dissect the internal anatomy of the Large Language Model.



Day 1 established the boundaries from basic predictive matrices to generative ecosystems.
Day 2 begins our deep dive: The Anatomy and Internal Workings of the LLM.